

Appendix

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Appendix A

Supplementary Tables

This appendix holds the material that supports Chapter 3 but was too long to include there. The notebook that produced every table and figure is provided separately as source code, so no code is reproduced here.

A.1 Full dataset specifications

Chapter 3 reports two matched covariates per dataset. Table A.1 gives the per-class means of all the properties we track, so that the effect of each ladder round can be checked directly. A property whose three class means are equal is matched and can no longer explain any separation between the classes.

dataset	round	class	$ E $	Δ	δ	triangles	clustering	diameter
edit-wide	R1	0	8.7	3.7	1.1	1.2	0.19	4.6
		1	17.7	6.1	2.4	12.4	0.51	3.0
		2	26.1	7.6	4.2	34.8	0.76	2.0
conn-wide	R1	1 comp.	12.9	4.9	1.6	6.2	0.35	3.2
		2 comp.	11.7	4.6	1.0	5.5	0.36	2.9
		3 comp.	11.0	4.4	0.6	5.1	0.38	2.5
conn-wide	R3	1 comp.	8.8	3.9	1.4	2.1	0.27	4.2
		2 comp.	8.8	3.9	1.4	2.3	0.30	2.8
		3 comp.	8.8	3.9	1.4	2.4	0.32	1.8
geom-wide	R1	diam. 2	13.2	7.1	2.0	8.4	0.42	2.0
		diam. 4	12.6	4.7	1.7	5.9	0.34	4.0
		diam. 6	10.0	3.5	1.2	2.1	0.22	6.0
geom-wide	R3	diam. 2	11.6	8.0	1.9	6.1	0.36	2.0
		diam. 4	11.6	4.0	1.9	4.8	0.31	4.0
		diam. 6	11.6	3.9	1.9	3.3	0.27	6.0
pure-label	R1	no B	10.8	4.5	1.5	4.1	0.31	2.7
		half B	10.8	4.5	1.5	4.1	0.31	2.7
		all B	10.8	4.5	1.5	4.1	0.31	2.7
colorability	R3	$\chi = 2$	9.6	4.1	1.4	0.0	0.00	3.1
		$\chi = 3$	9.6	4.2	1.4	1.8	0.24	2.9

Table A.1: Per-class means of the tracked properties. `pure-label` has identical values in all three classes because the three classes reuse the same graph structures and differ only in node labels. In `geom-wide` R3, maximum degree remains unequal because matching it in addition to edge count and minimum degree was infeasible at order nine.

Two designs could not be built as intended and are recorded here rather than silently dropped. Matching maximum degree in `geom-wide` R3, in addition to edge count and minimum degree, left no graphs at order nine. Building a `conn-wide` contrast with classes at one, three and five components, rather than one, two and three, left only four graphs per class after matching and was abandoned.

A.2 The full confound ladder

Chapter 3 shows selected rows. Table A.2 gives all 44 cells, including the R1 and R2 rounds omitted from the main text. “Baseline” is the best separation achievable by any single elementary property, and the column beside it names that property. The predicted verdict was generated before this experiment was run, from the results of Sections 3.3 and 3.4 only.

dataset	round / dist	AUC	null	baseline	by	ARI	predicted	observed
colorability	R1 GED	0.854	0.526	0.934	triangles	0.476	confounded	confounded
colorability	R1 Spectral	0.829	0.524	0.938	triangles	0.490	confounded	confounded
colorability	R1 SP	0.631	0.518	0.938	triangles	0.112	confounded	confounded
colorability	R1 WL	0.585	0.518	0.938	triangles	0.072	confounded	confounded
colorability	R2 GED	0.993	0.550	0.849	max. deg.	1.000	confounded	succeeds
colorability	R2 WL	0.904	0.541	0.847	max. deg.	0.375	confounded	succeeds
colorability	R2 Spectral	0.891	0.538	0.847	max. deg.	0.316	confounded	succeeds
colorability	R2 SP	0.839	0.546	0.847	max. deg.	0.448	confounded	confounded
colorability	R3 GED	0.597	0.522	0.862	clustering	0.002	blind	confounded
colorability	R3 Spectral	0.535	0.523	0.826	clustering	−0.011	blind	confounded
colorability	R3 SP	0.503	0.526	0.826	clustering	−0.013	blind	blind
colorability	R3 WL	0.503	0.521	0.826	clustering	−0.006	blind	blind
conn-wide	R1 SP	0.660	0.528	0.734	min. deg.	0.223	confounded	confounded
conn-wide	R1 Spectral	0.644	0.528	0.734	min. deg.	0.164	confounded	confounded
conn-wide	R1 WL	0.610	0.523	0.734	min. deg.	0.051	confounded	confounded
conn-wide	R1 GED	0.603	0.527	0.716	min. deg.	−0.024	confounded	confounded
conn-wide	R2 SP	0.708	0.534	0.745	min. deg.	0.234	succeeds	confounded
conn-wide	R2 WL	0.650	0.526	0.745	min. deg.	0.229	confounded	confounded
conn-wide	R2 GED	0.636	0.530	0.731	min. deg.	0.138	blind	confounded
conn-wide	R2 Spectral	0.559	0.530	0.745	min. deg.	0.000	confounded	confounded
conn-wide	R3 SP	0.950	0.535	0.783	diameter	0.780	succeeds	succeeds
conn-wide	R3 WL	0.611	0.531	0.783	diameter	0.079	confounded	confounded
conn-wide	R3 Spectral	0.560	0.524	0.783	diameter	0.015	confounded	confounded
conn-wide	R3 GED	0.551	0.527	0.757	diameter	0.048	confounded	confounded
edit-wide	R1 Spectral	1.000	0.539	0.983	triangles	1.000	confounded	succeeds
edit-wide	R1 SP	0.989	0.536	0.983	triangles	1.000	confounded	succeeds
edit-wide	R1 GED	0.943	0.540	0.992	triangles	0.867	confounded	confounded
edit-wide	R1 WL	0.901	0.531	0.983	triangles	0.848	confounded	confounded
geom-wide	R1 SP	0.986	0.536	0.792	max. deg.	0.983	confounded	succeeds
geom-wide	R1 Spectral	0.831	0.531	0.792	max. deg.	0.489	confounded	succeeds
geom-wide	R1 GED	0.676	0.536	0.799	max. deg.	0.228	confounded	confounded
geom-wide	R1 WL	0.635	0.521	0.792	max. deg.	0.074	confounded	confounded
geom-wide	R2 SP	0.979	0.539	0.875	max. deg.	1.000	succeeds	succeeds
geom-wide	R2 Spectral	0.806	0.533	0.875	max. deg.	0.506	confounded	confounded
geom-wide	R2 GED	0.797	0.533	0.852	max. deg.	0.248	blind	confounded
geom-wide	R2 WL	0.676	0.523	0.875	max. deg.	0.409	blind	confounded
geom-wide	R3 SP	0.972	0.539	0.895	max. deg.	0.889	succeeds	succeeds
geom-wide	R3 Spectral	0.807	0.531	0.895	max. deg.	0.273	confounded	confounded
geom-wide	R3 GED	0.795	0.529	0.880	max. deg.	0.185	blind	confounded
geom-wide	R3 WL	0.669	0.522	0.895	max. deg.	0.302	blind	confounded
pure-label	R1 WL	0.994	0.545	0.500	—	1.000	succeeds	succeeds
pure-label	R1 GED	0.916	0.536	0.500	—	0.423	succeeds	succeeds
pure-label	R1 SP	0.460	0.537	0.500	—	−0.041	blind	blind
pure-label	R1 Spectral	0.458	0.540	0.500	—	−0.048	blind	blind

Table A.2: All 44 cells of the prediction test. Twenty-six predictions are exactly correct. No cell predicted blind was observed to succeed, and none predicted to succeed was observed blind.

A.3 Grid size and shape

Table A.3 gives the per-size detail behind Section 3.9. Every grid position stays active at every size, so the number of prototypes is always the number requested.

K	4	6	9	12	16	20	25
active prototypes	4.0	6.0	9.0	12.0	16.0	20.0	25.0
quantization error	1.529	1.319	1.102	0.944	0.754	0.585	0.420
silhouette	0.371	0.391	0.409	0.423	0.426	0.428	0.434
clusters found	2.4	2.7	3.1	3.4	3.7	3.9	4.1
ARI	0.374	0.346	0.355	0.326	0.313	0.323	0.334

Table A.3: Behaviour against grid size, averaged over all cells. Quantization error falls monotonically in every one of the 132 cases tested; silhouette rises monotonically in 35% of them.

The grid size giving the best recovery is $K = 4$ in 21 of 44 cells, $K = 9$ in 7, $K = 6$ in 5, $K = 25$ in 5, $K = 20$ in 4 and $K = 12$ in 2.

For grid shape at fixed K , with the neighbourhood width scaled to the grid so that shape is not confounded with smoothing: at $K = 9$ the 3×3 grid gives an ARI of 0.332 against 0.245 for 1×9 ; at $K = 12$ the 3×4 grid gives 0.281 against 0.216 for 2×6 and 0.183 for 1×12 ; at $K = 16$ the 1×16 grid gives 0.373 against 0.323 for 4×4 and 0.204 for 2×8 . Shape accounts for roughly 3% of the variance in recovery at fixed K , with no consistent direction.

A.4 Claim ledger

Table A.4 lists every claim made in the thesis with the evidence behind it and how strongly we state it. “Established” means computed over a complete set of graphs or following from arithmetic. “Supported” means measured by more than one method, or by one carefully controlled experiment. “Refuted” means a hypothesis was stated in advance and the experiment contradicted it.

A.5 Reproducibility

All experiments were run in a single Python notebook, provided as source code alongside this thesis. Graphs are generated with **nauty** (**geng**), handled with **networkx**, and numerical work uses **numpy**, **scipy** and **scikit-learn**.

Distance matrices are computed once and cached on disk, keyed by the graph set and the distance, so re-running the notebook does not recompute them. Graph edit distance dominates the cost: a single matrix on forty-five graphs of order nine takes roughly two minutes, against under a second for the other three distances. For this reason it is computed on one data seed per dataset while the others use four, and it is excluded from the algorithm comparison entirely. The time budget is 0.15 seconds per pair.

Two random seeds are separated throughout. The data seed controls which graphs are drawn from the pool, four are used. The algorithm seed controls initialisation, three to five are used depending on the experiment. Permutation nulls use three hundred label shuffles and bootstrap intervals use five hundred resamples.

On a single processor core, the complete notebook takes roughly forty minutes with an empty cache and under ten minutes thereafter, with the first run dominated by the eleven graph edit distance matrices.

claim	section	evidence	strength
SP cannot distinguish 99.6% of 8-node graphs from another; largest group 840	3.2	complete enumeration	established
Cospectral collision reaches 13.5% at $n = 7$ and 14.5% at $n = 8$	3.2	exact characteristic polynomial	established
WL collapses 4.9–6.1% across orders six to eight	3.2	complete enumeration	established
Under unit costs GED returns 2.0000 for every rewire	3.4	arithmetic	established
Spectral is 95.4% predictable from seven elementary properties	3.3	regression, one population	supported
WL is 41.2% predictable; the rest is not any of the seven	3.3, 3.5, 3.8	regression, dimensionality, characterization	supported
No distance responds to one triangle at fixed edge count	3.4	paired single-edit test	supported
SP is specifically sensitive to diameter	3.3, 3.4, 3.7	three independent methods	supported
Recovery is governed by separation; threshold near 1.5 sd	3.5, 3.6	two dataset families	supported
Difficulty explains 77.6% of recovery, algorithm 0.9%	3.6	controlled comparison	supported
Seed selection explains 10.6%, eleven times the algorithm effect	3.6	swept seed rule	supported
GED is not Euclidean (negative mass 0.234)	3.5	two runs, different data	supported
Component count is encoded in the spectrum but inaccessible to the spectral distance	3.7	matched dataset	supported
Map concentration tracks recovery at +0.717	3.9	full grid sweep	supported
The layout is better than a random one by a factor of about 1.5	3.9	permuted-grid null	supported
Quantization error is unrelated to recovery (+0.053)	3.9	full grid sweep	supported
Silhouette reports a real partition, not a correct one	3.8	blindness control dataset	supported
No distance gives a repeatable clustering on undesigned graphs	3.8	seed and subsample agreement	supported
Chromatic number is invisible to all four under edge matching, while one scalar decodes it at 0.826	3.10	three-round ladder	supported
Predictions from the distance level transfer in ordering but not exactly (59%, no inversions)	3.7	mechanical prediction test	supported
Grid size can be chosen by a knee on quantization error	3.9	three procedures against fixed choices	refuted
Round 2 of the colorability ladder tests chromatic sensitivity	3.10	size confound, tracks max. degree	refuted

Table A.4: Claim ledger. Every statement in Chapter 4 corresponds to a row here.